An Evaluation of Massively Parallel Algorithms for DFA Minimization

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CERTIFICATE

This is to certify that the seminar report entitled **"An Evaluation of Massively Parallel Algorithms for DFA Minimization"** submitted by **Gokul Krishna (Reg. No.: MUT22CS066)** of Semester VII is a bonafide account of the work done by him under our supervision during the academic year 2024–2025.

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Contents

Abstract

The minimization of deterministic finite automata (DFA) is a fundamental problem with practical applications in model checking, compilers, and verification. This paper implements and compares several massively parallel GPU algorithms for DFA minimization, including a transitive-closure (NC) method and three partitionrefinement variants (naive leader-election, sorting-based, and a hybrid partitionrefinement augmented with a partial transitive closure). All methods were implemented in CUDA and evaluated on synthetic benchmark families (Fibonacci and bit-splitter automata) as well as real-world models from the VLTS suite. Experimental results show that the transitive-closure approach, despite its theoretical appeal, requires prohibitive resources and does not scale well on current GPU hardware; by contrast, partition-refinement methods are more practical. The sorting-based partitioning achieves the best performance on many VLTS instances by enabling aggressive block splitting, while the hybrid enhancement produces large speedups on automata with long uniform paths. Implementation details, runtime, iteration counts, and memory usage are reported, and practical guidelines for selecting an appropriate GPU minimization strategy based on input structure and alphabet size are presented.

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List of Abbreviations

AMD	Advanced Micro Devices
CAS	Compare-And-Swap
CPU	Central Processing Unit
CRCW	Concurrent Read Concurrent Write
CUDA	Compute Unified Device Architecture
CV	Computer Vision
DDR5	Double Data Rate 5 (memory standard)
DFA	Deterministic Finite Automaton
FLOPs	Floating Point Operations per Second
FP32	32-bit Floating Point
GPU	Graphics Processing Unit
GPGPU	General-Purpose GPU
INT8	8-bit Integer
NC	Nick's Class
NFA	Nondeterministic Finite Automaton
NVIDIA	NVIDIA Corporation
PRAM	Parallel Random Access Machine
PR	Partition Refinement
RAM	Random Access Memory
RE	Regular Expression
RTX	Ray Tracing Texel eXtreme (NVIDIA GPU series)
TC	Transitive Closure
TRANS	Transitive Closure-Based Algorithm
VLTS	Very Large Transition Systems
VRAM	Video Random Access Memory

Chapter 1

Introduction

1.1 Background and Relevance

Deterministic Finite Automata (DFA) are one of the oldest and most widely studied computational models in theoretical computer science. They provide a formal framework for representing and processing regular languages, which appear in numerous domains—from lexical analysis in compilers to communication protocol verification and digital circuit design. A DFA, being a deterministic state machine, transitions between a finite number of states according to an input alphabet, determining whether a given input string belongs to a particular language.

Despite their conceptual simplicity, DFAs play an integral role in many modern computing systems. They form the mathematical backbone of regular expression engines, lexical analyzers, formal verification tools, and even some artificial intelligence models. The ability to model complex behaviors using finite states, while ensuring deterministic outcomes, makes DFAs an indispensable component of computational theory and practice.

A critical aspect of automaton theory is **DFA minimization**, which refers to constructing an equivalent DFA with the smallest possible number of states. The minimized DFA recognizes exactly the same language as the original automaton but with a reduced state space. This reduction has profound implications:

- It improves computational efficiency during pattern matching and parsing.
- It minimizes memory consumption in embedded and real-time systems.
- It simplifies debugging, analysis, and visualization of models.

Historically, DFA minimization has been explored as a purely sequential problem. Moore's algorithm (1956) introduced the earliest formal procedure for minimizing DFAs using iterative refinement of equivalence classes. Later, Hopcroft's algorithm [2] improved efficiency to $O(n \log n)$, making it a long-standing benchmark for sequential performance. Paige and Tarjan [3] further extended the concept through their general partition refinement framework, applicable to a wider class of equivalence relations such as bisimulation. These algorithms remain foundational to automata theory and form the basis of modern compilers and verification tools.

However, computing paradigms have changed dramatically since the 1970s. Today's computational landscape is dominated by **parallel processing architectures**, where the focus has

shifted from single-core sequential performance to massive concurrency. Graphics Processing Units (GPUs), originally designed for rendering graphics, now serve as high-throughput parallel processors capable of performing billions of operations per second. With frameworks such as NVIDIA CUDA and OpenCL, GPUs have become mainstream platforms for general-purpose parallel computation.

This evolution in hardware architecture has prompted renewed interest in how traditional automata-theoretic algorithms can be adapted to leverage such massive parallelism. DFA minimization, being both computationally intensive and structurally regular, serves as an ideal case study for exploring this question. While its underlying theory remains the same, its implementation can now exploit thousands of concurrent threads to process state transitions and equivalence computations simultaneously.

Recent research has begun exploring this new frontier. Works like that of Tewari, Ravikumar, and Arora [4] established that DFA minimization theoretically belongs to Nick's Class (NC), meaning it can be efficiently parallelized using a polynomial number of processors. Later, Wijs [5] demonstrated GPU-based bisimilarity checking, providing concrete evidence that automata equivalence computations can indeed benefit from GPU architectures. Building on these foundations, Martens and Wijs [1] provided a comprehensive evaluation of massively parallel algorithms for DFA minimization, benchmarking several GPU-optimized methods and analyzing their performance across diverse automata families.

Their findings revealed that theoretical parallelism does not automatically translate to real-world efficiency. Although closure-based algorithms offer superior theoretical complexity, they often require excessive memory and synchronization overheads, making them impractical for large-scale GPUs. Conversely, algorithms based on partition refinement—though theoretically less optimal—tend to perform better in real GPU environments due to their lightweight data dependencies and regular memory access patterns.

Therefore, the motivation for this research lies in bridging the gap between **parallel complexity theory and GPU-based practical implementation**. The study investigates how DFA minimization can be reformulated for GPUs, aiming to determine which algorithmic strategies achieve genuine scalability when mapped onto massively parallel architectures.

1.1.1 Significance of DFA Minimization in Modern Systems

The relevance of DFA minimization extends well beyond theoretical computer science. In compiler design, minimized DFAs ensure that lexical analyzers operate with minimal lookup time. In communication systems, minimized automata enable faster protocol conformance checking. In model checking, where systems are represented as large state machines, reducing the automaton size can drastically improve verification time and memory efficiency.

In contemporary computing, DFA-based models also appear in network intrusion detection systems, string-matching engines, and hardware synthesis tools. These applications often operate at large scales—processing millions of transitions per second—where even modest reductions in automaton size translate into significant performance gains. The push toward parallel architectures makes it imperative that minimization algorithms scale proportionally with available hardware resources.

1.1.2 Transition from Sequential to Parallel Paradigms

Traditional DFA minimization algorithms such as those by Hopcroft and Paige–Tarjan were designed for serial execution, relying heavily on ordered refinement of partitions. These algorithms exhibit dependencies that are difficult to parallelize directly. For instance, the outcome of one partition refinement iteration often determines the input for the next, introducing strict sequential ordering.

With the emergence of parallel models such as PRAM (Parallel Random Access Machine), researchers began exploring how DFA minimization could theoretically exploit parallelism. Tewari et al. [4] showed that through transitive closure computations on state-pair graphs, DFA minimization could, in theory, be achieved in polylogarithmic time using polynomial processors. Yet, these results were largely theoretical, as PRAM's assumptions of uniform memory access and infinite processor availability do not hold in real hardware environments.

The arrival of GPUs offered a tangible platform for parallel computation. Modern GPUs consist of thousands of lightweight cores organized into streaming multiprocessors, supporting simultaneous execution of numerous threads. Unlike CPUs, which optimize for latency, GPUs are optimized for throughput—making them ideal for problems involving large, data-parallel workloads such as automata processing.

This shift from theoretical to practical parallelism brings new challenges. GPUs impose strict constraints on memory access patterns, requiring coalesced reads and writes to achieve full efficiency. Synchronization across threads is limited, and shared memory size is restricted. Therefore, adapting DFA minimization for GPUs is not merely a matter of parallelizing existing algorithms—it requires redesigning data structures, memory layouts, and synchronization strategies to align with the GPU execution model.

1.2 Problem Statement

Although DFA minimization has long been considered theoretically parallelizable, achieving efficient performance on physical hardware remains a complex challenge. Algorithms with superior asymptotic complexity—such as transitive-closure-based methods—often exhibit high memory overhead and limited scalability on GPUs. In contrast, algorithms with less impressive theoretical performance, such as partition refinement methods, frequently deliver better practical results.

The central problem addressed in this work can be articulated as follows:

> *How can deterministic finite automaton minimization algorithms be reformulated and optimized to exploit modern GPU architectures while balancing theoretical efficiency with practical scalability?*

This research seeks to investigate the discrepancy between theoretical parallel performance and actual GPU execution efficiency. Despite belonging to NC, the DFA minimization problem continues to face several practical barriers:

1. **Memory Overhead:** Closure-based algorithms require quadratic space in the number of states, exceeding the capacity of typical GPUs.

2. **Synchronization Bottlenecks:** Parallel refinement operations often require atomic updates or barriers, leading to contention.

3. **Irregular Data Access Patterns:** The inherently irregular structure of automata causes uncoalesced memory accesses, reducing throughput.

4. **Trade-Off Between Iterations and Cost:** Reducing iteration counts via more complex refinement methods can increase per-iteration computational load.

Furthermore, certain automata families—such as Fibonacci or Bit-Splitter DFAs—expose weaknesses in specific algorithms by triggering worst-case partition refinements. As shown in Martens and Wijs [1], such input-sensitive behavior leads to unpredictable runtime scaling even on powerful GPU hardware.

Therefore, the need arises to systematically evaluate different parallel minimization techniques under controlled GPU conditions. By implementing, profiling, and analyzing multiple algorithmic approaches, this study aims to identify which strategies best translate theoretical parallel efficiency into real-world performance gains.

1.2.1 Research Questions

The investigation revolves around three key research questions:

1. Why do algorithms with theoretically optimal complexity fail to achieve practical scalability on GPUs?

2. Among partition refinement algorithms, which variant—naive, sorting-based, or hybrid—delivers the best balance between iteration cost and convergence rate?

3. Can partial transitive closure integration accelerate convergence without overwhelming GPU memory resources?

Answering these questions provides not only insight into the specific case of DFA minimization but also a broader understanding of how to adapt complex graph-based algorithms to GPU architectures efficiently.

1.3 Objectives

The primary aim of this study is to design, implement, and evaluate massively parallel DFA minimization algorithms optimized for GPU execution. This involves both theoretical and experimental components that collectively contribute to bridging the gap between algorithmic theory and hardware-level performance.

1.3.1 General Objective

To analyze and optimize deterministic finite automaton minimization techniques for GPU architectures, achieving scalable, memory-efficient, and high-throughput implementations.

1.3.2 Specific Objectives

1. **Algorithmic Implementation:** Develop GPU-based implementations of multiple DFA minimization algorithms—including transitive-closure-based and partition-refinement-based approaches—using CUDA C++ as the programming framework.

2. **Experimental Evaluation:** Benchmark these algorithms using well-known automata families such as Fibonacci, Bit-Splitter, and VLTS, measuring runtime, scalability, and memory utilization.

3. **Performance Analysis:** Analyze the relationship between theoretical algorithmic complexity and empirical GPU performance, identifying discrepancies between asymptotic predictions and real outcomes.

4. **Hybrid Optimization:** Explore hybrid approaches that combine partial closure propagation with partition refinement to accelerate convergence while maintaining manageable resource consumption.

5. **Practical Contribution:** Provide actionable insights for designing GPU-oriented automata algorithms that balance simplicity, scalability, and efficiency.

1.3.3 Scope and Expected Contributions

This research situates itself at the intersection of formal language theory, algorithm design, and high-performance computing. While classical DFA minimization has been extensively studied in sequential contexts, its parallel adaptation remains relatively underexplored. By building upon recent GPU-based frameworks [5, 1], this work contributes to:

- Establishing a unified framework for evaluating multiple minimization algorithms under comparable GPU conditions.
- Identifying algorithmic structures that best align with GPU memory and concurrency models.
- Providing empirical results that inform the design of future automata-processing systems.

Beyond theoretical analysis, the study aims to contribute practically to the fields of compiler optimization, model checking, and software verification, where automata-based computations are frequent and performance-sensitive.

1.3.4 Broader Implications

The implications of this research extend beyond DFA minimization. Many computational problems—such as bisimulation reduction, equivalence checking, and symbolic state-space exploration—share structural similarities with DFA minimization. Insights gained from this study could inform optimization techniques in these related areas as well.

In a broader sense, the work exemplifies how classic theoretical algorithms can be revitalized through modern parallel architectures. As GPUs continue to evolve with larger memory capacities and improved synchronization primitives, reimagining core algorithms for such architectures is essential to fully harness their potential.

1.4 Chapter Summary

In summary, this chapter introduced the motivation and context for the research on GPU-based DFA minimization. It discussed the historical development of DFA minimization algorithms, the advent of parallel computing, and the growing importance of GPUs in general-purpose computation. It outlined the core challenges involved in adapting theoretical algorithms to real hardware and formulated the central problem statement and objectives guiding this study.

The next chapters build upon this foundation. Chapter 2 provides an in-depth review of existing work in sequential, parallel, and GPU-based DFA minimization. Chapter 3 then presents the detailed methodology, system design, and implementation strategies developed to investigate and evaluate these algorithms.

Chapter 2

Literature Survey

2.1 Overview of Existing Techniques

Minimization of Deterministic Finite Automata (DFA) is a classic optimization problem that has attracted continuous research attention since the mid-twentieth century. The task of DFA minimization is to construct an automaton that accepts the same language as the given DFA but with the smallest possible number of states. Such reduction has direct implications in compiler design, lexical analysis, pattern recognition, and formal verification tools, where execution speed and memory footprint are paramount.

From the earliest sequential approaches to the more recent GPU-accelerated techniques, the problem has undergone a profound transformation in both theoretical understanding and practical implementation. The earliest methods were rooted purely in algorithmic logic and mathematical reasoning, whereas contemporary techniques combine insights from automata theory with high-performance computing and memory optimization strategies.

Moore's algorithm (1956) laid the first formal foundation for the minimization process through an iterative partitioning mechanism that refines the state space until all distinguishable states are separated. However, its $O(n^2)$ complexity limited scalability for large automata. Hopcroft's algorithm [2] later revolutionized the field by introducing a more efficient refinement procedure that reduced the complexity to $O(n \log n)$. This improvement stemmed from a sophisticated use of partition refinement that avoided redundant recalculations, effectively optimizing the sequence of state distinctions.

The evolution continued with Paige and Tarjan's general partition refinement framework [3], which not only enhanced computational efficiency but also generalized the concept beyond DFAs to handle relational systems such as bisimulation and coarsest-partition problems in process algebra. Their work bridged automata theory and graph theory, establishing a unifying principle for equivalence computation.

The advent of parallel computing opened new dimensions for DFA minimization. The work by Tewari, Ravikumar, and Arora [4] demonstrated that minimization could theoretically be performed efficiently in parallel under the PRAM (Parallel Random Access Machine) model, categorizing it within Nick's Class (NC). This discovery confirmed that DFA minimization is amenable to parallelism, although implementing such models on physical hardware remained challenging.

The subsequent decade witnessed a shift toward exploiting graphics processing units (GPUs) as general-purpose parallel processors. The contribution of Wijs [5] was particularly notable in demonstrating how bisimulation and equivalence checking — problems structurally analogous to DFA minimization — can be accelerated on GPUs. This approach introduced practical design patterns such as parallel partition updates, coalesced memory accesses, and the use of prefix sums for efficient partition refinement.

Finally, the work of Martens and Wijs [1] represents a culmination of these efforts, offering a comparative evaluation of massively parallel DFA minimization algorithms. Their results revealed the intricate trade-off between theoretical complexity and real-world GPU constraints such as memory coalescing, synchronization latency, and warp divergence. Collectively, these studies trace a clear intellectual lineage from purely sequential theory to hardware-aware parallel execution.

2.2 Key Studies and Methodologies

2.2.1 Hopcroft's $O(n \log n)$ Minimization Algorithm (1971)

Methodology: Hopcroft's minimization algorithm [2] improved upon Moore's earlier quadratic method by refining the notion of partition management. It begins by separating all states into two disjoint sets: accepting and non-accepting states. Then, through iterative refinement, the algorithm repeatedly splits these partitions based on how states respond to input symbols. When two states yield different successor partitions under a particular symbol, they are separated into distinct blocks.

The critical innovation lies in the careful selection of the smallest active partition as a splitter. This ensures that, over the algorithm's execution, each state participates in only a logarithmic number of splits relative to the total number of states. Consequently, the algorithm achieves an asymptotic time complexity of $O(n \log n)$, representing a near-optimal result for sequential minimization.

In addition to its efficiency, Hopcroft's algorithm introduced a modular design that facilitated practical implementations. By representing partitions as linked structures and maintaining transition mappings, the algorithm effectively reduced redundant comparisons between large state sets. Its predictable data flow and low memory requirements made it ideal for integration into compiler design tools and lexical analyzers.

Strengths:

- Achieves theoretical optimality for sequential minimization.
- Data structures are compact and predictable in runtime behavior.
- Serves as the canonical baseline for all subsequent improvements.

Weaknesses:

- The algorithm is inherently sequential; each refinement depends on the previous.
- Inefficient for automata with skewed transition distributions.

- Performance deteriorates for DFAs with large alphabets or unbalanced structures.

Hopcroft's approach remains a cornerstone in automata theory, often reimplemented as a serial reference point in comparative evaluations of modern parallel techniques.

2.2.2 Partition Refinement and Paige–Tarjan Framework (1987)

Methodology: The work of Paige and Tarjan [3] marked a paradigm shift by abstracting partition refinement into a generalized mathematical framework. Their method efficiently computes the coarsest stable partition of a relational structure. Instead of focusing solely on deterministic transitions, their algorithm operates on generic relations and efficiently refines equivalence classes through localized updates.

The algorithm introduces dependency tracking between partitions, ensuring that when a block is refined, only the relevant dependent blocks are reconsidered. This avoids unnecessary iterations and reduces total runtime. On sparse graphs and automata with few transitions per state, it performs nearly linearly with respect to the number of edges.

Strengths:

- Extends minimization to a broader class of equivalence problems, including bisimulation.
- Highly efficient for sparse state-transition systems.
- Enables incremental refinement with early termination.

Weaknesses:

- Complex dependency maintenance increases implementation difficulty.
- The approach, while efficient, remains largely sequential.
- Synchronization overheads appear when parallelized naively.

Relevance to Modern Systems: Many GPU-based algorithms derive directly from Paige–Tarjan's model. By representing partitions as contiguous arrays and using parallel primitives (such as sort and scan operations), researchers have effectively adapted this framework to parallel hardware, translating theoretical stability guarantees into practical parallel performance.

2.2.3 Parallel DFA Minimization (Tewari, Ravikumar & Arora, 2002)

Methodology: Tewari, Ravikumar, and Arora [4] investigated the DFA minimization problem under the lens of parallel computation. Using the PRAM model, they represented the DFA as a two-dimensional matrix of state pairs, allowing concurrent processing of distinguishability information. Parallel transitive closure algorithms were then applied to compute the equivalence relation across all state pairs.

Their study established that DFA minimization can theoretically be completed in $O(\log^2 n)$ time using $O(n^2)$ processors, proving that the problem lies within Nick's Class (NC). This classification implies that DFA minimization is highly parallelizable, at least in principle, even though such a processor count is impractical for real systems.

Strengths:

- Provided the first formal parallel complexity analysis of the minimization problem.
- Demonstrated the feasibility of logarithmic parallel depth.
- Introduced the use of transitive closure as a parallel primitive.

Weaknesses:

- Requires quadratic memory, making it unsuitable for large automata.
- Synchronization across processors introduces non-trivial delays.
- Assumes idealized PRAM conditions that do not map directly to real hardware.

Despite these limitations, this work remains foundational. The decomposition of the minimization process into transitive closure and refinement steps paved the way for hybrid GPU-based methods that combine both strategies efficiently.

2.2.4 GPU-Accelerated Bisimilarity Checking (Wijs, 2015)

Methodology: Wijs [5] presented a breakthrough in bringing automata equivalence computation to GPUs. By focusing on bisimilarity checking — a problem structurally similar to DFA minimization — the research demonstrated how parallel hardware can accelerate state-space partitioning.

The algorithm exploits GPU architecture by representing automata using compressed sparse row (CSR) structures, minimizing memory overhead and enabling coalesced memory access. Each GPU thread processes a subset of transitions, computing equivalence signatures that determine partition refinement. Key operations such as sorting, prefix sums, and reductions are implemented using parallel primitives, ensuring thousands of threads contribute concurrently.

Strengths:

- Demonstrated that fine-grained partition refinement can scale across thousands of cores.
- Achieved significant runtime reductions for large automata.
- Established design principles for GPU data layout and synchronization.

Weaknesses:

- Dependent on GPU memory limits; extremely large DFAs can exceed capacity.
- Load imbalance across threads leads to underutilization on irregular datasets.
- Atomics and synchronization can become bottlenecks in dense graphs.

Relevance: Wijs' framework laid the groundwork for later GPU-based DFA minimization approaches, demonstrating that partition-based algorithms can achieve practical acceleration while preserving theoretical correctness.

2.2.5 Modern Parallel and Hybrid Approaches (Martens & Wijs, 2024)

Methodology: The recent study by Martens and Wijs [1] offers a systematic evaluation of massively parallel DFA minimization algorithms. Three major categories are compared:

- **Transitive-closure-based methods** — high theoretical parallelism but large memory footprint.
- **Partition-refinement-based methods** — moderate theoretical depth but highly efficient in practice.
- **Hybrid approaches** — combining both paradigms for optimal trade-off between iterations and memory.

Benchmarks were conducted on large automata families, including Fibonacci, Bit-Splitter, and VLTS datasets. The results indicated that partition-refinement methods consistently outperformed transitive-closure approaches on GPUs, primarily due to their reduced memory usage and improved cache locality.

Findings: Hybrid models that integrate partial closure computations with refinement steps show strong scalability, adapting well to different input structures. However, parameter tuning — such as determining closure depth — is crucial for achieving balanced performance.

Analysis: The study's results provide valuable empirical evidence for the transition from theoretical PRAM models to practical GPU realizations. By quantifying performance metrics such as kernel execution time, thread occupancy, and memory efficiency, Martens and Wijs effectively connect algorithmic structure with hardware behavior, a perspective absent from earlier theoretical research.

2.3 Evolution of Parallel DFA Minimization

The trajectory of DFA minimization research can be divided into three interconnected stages:

1. **Sequential Foundation:** Beginning with Moore and formalized by Hopcroft, this era focused on establishing asymptotically optimal algorithms within the constraints of single-core computation.
2. **Parallel Theoretical Exploration:** Driven by the rise of PRAM models, researchers such as Tewari et al. [4] investigated the inherent parallelism of minimization, revealing its NC classification.
3. **GPU-Enabled Realization:** Modern implementations by Wijs [5] and Martens & Wijs [1] demonstrate practical parallelization through CUDA and GPU-aware algorithm design.

This evolution illustrates an enduring theme in computer science: theoretical algorithms achieve their full potential only when re-engineered to align with contemporary hardware constraints. GPU-based solutions embody this synthesis by translating mathematical abstraction into data-parallel execution strategies.

2.4 Summary of Limitations

Although modern research has achieved remarkable progress, several recurring challenges persist:

- **Resource Blow-Up:** Algorithms relying on transitive closure often incur quadratic or cubic space complexity, limiting scalability on GPUs.
- **Input Sensitivity:** Certain automata families, such as Fibonacci or bit-splitter DFAs, exacerbate partition splitting and degrade average-case performance.
- **GPU Architectural Constraints:** Limited VRAM, non-uniform memory access patterns, and warp divergence affect performance consistency.
- **Iteration vs. Cost Trade-Off:** Signature-based methods reduce iterations but demand more computation per pass, leading to variable efficiency.
- **Synchronization Bottlenecks:** Heavy use of atomic operations can serialize threads, counteracting parallel speedup.

These limitations highlight that despite theoretical tractability, efficient parallel minimization still requires careful engineering to balance algorithmic elegance with architectural realities.

2.5 Conclusion

The study of DFA minimization reflects the broader evolution of algorithmic optimization under changing computational paradigms. The transition from Hopcroft's sequential $O(n \log n)$ algorithm [2] to hybrid GPU-accelerated approaches [1] represents a synthesis of classical automata theory with high-performance computing.

Hopcroft's and Paige–Tarjan's frameworks provided mathematical rigor and asymptotic efficiency, while Tewari et al. [4] extended these insights into parallel theory. Building upon these foundations, Wijs [5] and later Martens & Wijs [1] translated these ideas into concrete, hardware-optimized systems capable of processing vast state spaces within practical time frames.

Future directions include exploring distributed GPU clusters, adaptive hybridization strategies, and out-of-core memory management to overcome scalability bottlenecks. As computational platforms continue to evolve, DFA minimization remains a vibrant intersection of formal theory and practical systems engineering, exemplifying how foundational ideas can persist and adapt across technological generations.

Chapter 3

Proposed Methodology

3.1 System Architecture Overview

The proposed methodology builds upon the GPU-oriented framework described by Martens and Wijs [1], which investigates the practical realization of massively parallel algorithms for deterministic finite automaton (DFA) minimization. The framework is designed to evaluate multiple algorithmic paradigms—each derived from well-established theoretical models—and adapt them to exploit the computational capabilities of modern graphics processing units (GPUs).

The primary objective of this methodology is to reconcile the gap between theoretical parallelism, established in classical works such as Hopcroft [2] and Paige–Tarjan [3], and real-world GPU execution efficiency. Earlier PRAM-based theoretical models, such as those explored by Tewari et al. [4], established that DFA minimization belongs to Nick's Class (NC), but did not provide implementable solutions that scale on physical hardware. The present framework seeks to bridge this divide through a unified GPU-based architecture capable of executing and comparing multiple parallel minimization strategies within a controlled experimental setup.

The framework evaluates three algorithmic families:

1. **Transitive Closure-Based Minimization (TRANS)** — A theoretically complete, graph-oriented approach that models equivalence computation as a closure problem.

2. **Partition Refinement Algorithms (NaivePR, SortPR)** — Algorithms that iteratively split state partitions based on transition behavior, derived from Hopcroft and Paige–Tarjan's frameworks.

3. **Hybrid Transitive-Partition Minimization (TransPR)** — An adaptive approach combining the transitive closure model with partition refinement to balance iteration depth and per-pass computation.

Each algorithm follows a consistent input–output pipeline, differing primarily in how equivalence between states is computed and refined. By standardizing input formats, GPU kernel configurations, and validation stages, the architecture ensures fair cross-algorithm benchmarking and modular extensibility.

3.1.1 Workflow Structure

The methodology's overall workflow comprises four major phases, each optimized for concurrent GPU execution:

- **Input and Preprocessing:** The DFA is parsed and transformed into GPU-compatible data structures, using compact, contiguous representations that maximize memory coalescing.
- **Parallel Minimization:** One of the four core minimization algorithms (TRANS, NaivePR, SortPR, or TransPR) is executed using multiple CUDA kernels configured for data-parallel execution.
- **Refinement and Convergence Detection:** Partition updates are performed iteratively until stabilization, with convergence checked using GPU-wide reduction kernels.
- **Output Generation and Validation:** The minimized automaton is reconstructed from the final partition map and cross-verified for equivalence with the original input automaton.

This modular decomposition allows different algorithmic strategies to share the same execution environment, ensuring consistent memory handling, data access patterns, and timing metrics. The system design emphasizes minimal kernel-level synchronization and high warp occupancy to achieve near-optimal throughput.

3.2 Input Representation and Preprocessing

The preprocessing stage converts a deterministic finite automaton (DFA) into a flat, GPU-compatible representation. Given a DFA defined as a 5-tuple $(Q, \Sigma, \delta, q_0, F)$, where:

- Q is the finite set of states,
- Σ is the input alphabet,
- $\delta : Q \times \Sigma \to Q$ is the transition function,
- q_0 is the initial state, and
- $F \subseteq Q$ is the set of accepting states,

the goal is to encode δ and F in memory structures conducive to coalesced GPU memory access and efficient thread mapping.

3.2.1 Data Transformation

Traditional CPU representations of DFAs employ pointer-based adjacency lists or transition matrices. However, pointer chasing and irregular access patterns degrade GPU performance. Therefore, all DFA transitions are flattened into contiguous arrays:

$$\texttt{transitions}[i] = (\texttt{source}_i, \texttt{symbol}_i, \texttt{destination}_i)$$

This representation guarantees unit-stride memory access and facilitates thread-level parallelism.

State Encoding: Each DFA state is assigned a unique integer ID in $[0, |Q| - 1]$, permitting constant-time indexing of transitions. This enables straightforward mapping between transition entries and partition labels without branching logic.

Transition Compression: Transitions are stored in three parallel arrays (source, symbol, destination), minimizing redundancy. Depending on alphabet size $|\Sigma|$, compressed encoding techniques (e.g., bit-packing for small alphabets) further reduce memory usage.

Initial Partitioning: States are initially grouped into two blocks — accepting and non-accepting — forming the base partition for refinement algorithms. This structure is encoded as a `partition[]` array storing block identifiers for each state.

3.2.2 GPU Memory Preparation

To ensure efficient execution:

- Transitions are aligned to 128-byte boundaries to match GPU cache lines.
- Partition and transition arrays are stored in global memory, while block metadata resides in shared memory for frequent updates.
- Precomputed offsets and index ranges are maintained for each alphabet symbol to enable symbol-specific kernel launches.

This preprocessing step reduces inter-thread memory contention and lays the groundwork for scalable partition updates across thousands of parallel threads.

3.3 Core Minimization Algorithms

The framework implements four core algorithms, each targeting different aspects of DFA equivalence computation. These algorithms share input and output formats but differ in intermediate computation methods and GPU optimization strategies.

3.3.1 TRANS — Transitive Closure-Based Minimization

Overview: The TRANS algorithm models DFA minimization as a graph reachability problem. It constructs a *pair graph*, where each node represents a pair of DFA states (p, q). If two states are distinguishable under some input sequence, the pair (p, q) is marked as non-equivalent, and this distinction is propagated through closure computations.

Algorithmic Flow:

1. Construct a matrix representing all state pairs (p, q).
2. Initially mark pairs where one state is accepting and the other is not.
3. Iteratively propagate distinguishability: if $(\delta(p, a), \delta(q, a))$ is marked, then mark (p, q) as well.

4. Continue until no new pairs are marked—indicating closure stabilization.

Parallelization: On GPUs, each thread is assigned one or more pairs (p, q). Threads evaluate transitions in parallel and perform closure updates using shared bitmaps to represent distinguishability. The propagation phase is implemented as an iterative kernel launch where each pass updates the distinguishability matrix via bitwise OR operations.

Advantages:

- Conceptually simple and mathematically complete.
- Highly parallelizable—each pair computation is independent.

Limitations:

- Quadratic memory complexity $O(|Q|^2)$ restricts scalability.
- Iterative closure computation can result in many synchronization rounds.

Algorithm 1 Transitive DFA minimization `trans`.

Input: A DFA $A = (Q, \Sigma, \delta, F, q_0)$ where $|Q| = n$
Output: The minimal quotient automaton represented in the matrix *Apart*

1: $V :: Q \times Q$ ▷ Nodes of graph consisting of pair of states
2: *Apart* :: Array$[n^2]$ of type $\mathbb{B}$
3: *Reach* :: Array$[n^2][n^2]$ of type $\mathbb{B}$ ▷ Represents reachability in V, initially `false`
4: **do in parallel for** $(q, q') \in V$ ▷ Initializes data structures in parallel.
5: *Apart*$[(q, q')] := (q \in F \iff q' \notin F)$ ▷ State initially unequal
6: **for all** $a \in \Sigma$ **do**
7: *Reach*$[(q, q')][(\delta(q, a), \delta(q', a))] :=$ `true`
8: *stable* := `false`
9: **while** $\neg$*stable* **do**
10: *stable* := `true`
11: **do in parallel for** $s, t, u \in V$
12: **if** *Reach*$[s][t]$ and *Reach*$[t][u]$ and $\neg$*Reach*$[s][u]$ **then**
13: *Reach*$[s][u] :=$ `true`
14: **do in parallel for** $s, t \in V$
15: **if** *Reach*$[s][t]$ and *Apart*$[t]$ and $\neg$*Apart*$[s]$ **then**
16: *Apart*$[s] :=$ `true`
17: *stable* := `false`

Figure 3.1: Algorithm 1: Transitive DFA minimization (trans)

3.3.2 NaivePR — Baseline Partition Refinement Algorithm

Overview: The NaivePR algorithm is a straightforward GPU adaptation of Paige–Tarjan-style refinement. It partitions the state set and repeatedly refines blocks until convergence. This method replaces sequential refinement queues with parallel signature computations, assigning one thread per state.

Algorithmic Flow:

1. Initialize two blocks: accepting and non-accepting states.

2. Compute for each state a *signature*, defined as the vector of destination block IDs for each input symbol.

3. Sort states by signature; states sharing identical signatures form the same new block.

4. Repeat until partitions remain unchanged.

Leader Selection Variants:

- *Non-atomic (two-phase):* Leaders are elected in a first pass, and new block IDs are assigned in a second.

- *Atomic (single-phase):* Compare-and-swap (CAS) atomics are used to claim block leadership immediately.

The atomic variant reduces synchronization rounds but may introduce contention under heavy concurrency.

Parallel Implementation: Each thread independently computes transition signatures. GPU sorting primitives such as radix sort are employed to group identical signatures efficiently. Reduction kernels detect stable partitions by comparing old and new block IDs.

Advantages:

- Minimal control-flow divergence.

- Easily extensible to various DFA families.

Limitations:

- High sorting overhead for large state sets.

- Requires multiple refinement iterations to reach stability.

Algorithm 2 Parallel leader-election-based algorithm `naivePR`.

Input: A DFA $A = (Q, \Sigma, \delta, F, q_0)$ where $|Q| = n$
Output: The minimal quotient automaton represented in the array *block*

```
1:  block :: Array[n] of type Q
2:  new_leader :: Array[n] of type Q
3:  Select initial leader states q_f ∈ F and q_n ∈ Q \ F
4:  do in parallel for q ∈ Q
5:      block[q] := (q ∈ F ? q_f : q_n)                          ▷ Initialize
6:  stable := false
7:  while ¬stable do
8:      stable := true
9:      do in parallel for q,a ∈ Q × Σ
10:         if block[δ(q,a)] ≠ block[δ(block[q],a)] then
11:             new_leader[block[q]] := q                        ▷ Leader election
12:     do in parallel for q,a ∈ Q × Σ
13:         if block[δ(q,a)] ≠ block[δ(block[q],a)] then
14:             block[q] := new_leader[block[q]]                 ▷ Split from leader
15:             stable := false
```

Algorithm 3 Parallel leader-election based `naivePR` with atomics.

Input: A DFA $A = (Q, \Sigma, \delta, F, q_0)$, where $|Q| = n$
Output: The minimal quotient automaton represented in the array *block*

```
1:  block :: Array[n] of type Q
2:  new_block :: Q
3:  leader :: Q
4:  new_leader :: Array[n] of type Q
5:  Select initial leader states q_f ∈ F and q_n ∈ Q \ F
6:  do in parallel for q ∈ Q
7:      new_leader[q] := ⊥                                       ▷ Initialize
8:      block[q] := (q ∈ F ? q_f : q_n)
9:  while ¬stable do
10:     stable := true
11:     do in parallel for q,a ∈ Q × Σ
12:         leader := block[q]
13:         if block[δ(q,a)] ≠ block[δ(leader,a)] then
14:             {new_block := new_leader[leader];                ▷ Leader election with CAS
15:             new_leader[leader] = ⊥ ? new_leader[leader] := q}
16:             block[q] := new_block = ⊥ ? q : new_block        ▷ Split from leader
17:             stable := false
18:     do in parallel for q ∈ Q
19:         new_leader[q] := ⊥
```

Figure 3.2: Algorithms 2 and 3: NaivePR leader-election variants (non-atomic and atomic)

3.3.3 SortPR — Optimized Signature-Based Partition Refinement

Overview: SortPR refines the NaivePR method by employing GPU sorting and hashing optimizations. It minimizes redundant comparisons between states through precomputed signature vectors and hash-based compression.

Algorithmic Flow:

1. Label each transition with its source state and target block ID.
2. Assemble signature vectors for each state.
3. Apply parallel radix sort to group identical signatures.
4. Merge adjacent states with equal signatures into a new block.
5. Iterate until partitions stabilize.

GPU Implementation: SortPR uses shared memory for signature caching and warp-synchronous sorting to minimize divergence. Hash compression ensures signatures fit within per-thread memory limits. Prefix-sum scans efficiently identify block boundaries, eliminating serial bottlenecks.

Advantages:

- Outperforms NaivePR by reducing redundant recomputation.
- Achieves high GPU occupancy and low synchronization cost.

Limitations:

- Memory usage increases with large alphabets due to signature vectors.
- Sorting dominates runtime for small automata.

Algorithm 4 Parallel sorting-based algorithm `sortPR`

Input: A DFA $A = (Q, \Sigma, \delta, F, q_0)$, where $|Q| = n$ and $|\Sigma| = k$
Output: The minimal quotient automaton represented in the array *block*

```
1:  block :: Array[n] of type ℕ
2:  state :: Array[n] of type Q
3:  signature :: Array[n][k] of type Q
4:  new_block :: Array[n] of type ℕ
5:  num_blocks := 2
6:  do in parallel for q ∈ Q
7:      block[q] := (q ∈ F ? 0 : 1)                          ▷ Initialize
8:      state[q] := q
9:  repeat
10:     num_blocks := new_block[n − 1] + 1                   ▷ Number of blocks before iteration
11:     do in parallel for q, a ∈ Q × Σ
12:         signature[q][a] := block[δ(q, a)]
13:     sort(state, COMPARE)
14:     new_block := adjacent_diff(state, ARE_NEQ)           ▷ Place 1 for each change
15:     new_block[0] := 0
16:     new_block := inclusive_scan(new_block)               ▷ Compute new block labels
17:     do in parallel for q ∈ Q
18:         block[state[q]] = new_block[q]
19: until new_block[n − 1] + 1 = num_blocks

20: function COMPARE(q1, q2)
21:     if block[q1] > block[q2] then return false
22:     if block[q1] < block[q2] then return true
23:     for all a ∈ Σ do
24:         if signature[q1][a] > signature[q2][a] then return false
25:         if signature[q1][a] < signature[q2][a] then return true
26:     return false

27: function ARE_NEQ(q1, q2)
28:     if block[q1] ≠ block[q2] then return true
29:     for all a ∈ Σ do
30:         if signature[q1][a] ≠ signature[q2][a] then return true
31:     return false
```

Figure 3.3: Algorithm 4: Parallel sorting-based algorithm (sortPR)

3.3.4 TransPR — Hybrid Transitive-Partition Algorithm

Overview: The TransPR algorithm merges the strengths of both transitive closure and partition refinement approaches. It performs partial closure computations within each iteration to reduce redundant splits, striking a balance between computational depth and per-pass workload.

Algorithmic Flow:

1. Begin with partition refinement as in SortPR.

2. After each iteration, compute closure only on recently modified partitions.

3. Use closure results to preemptively refine blocks in subsequent iterations.

4. Alternate between refinement and closure until convergence.

Parallel Implementation: The hybrid design launches separate kernels for closure propagation and partition refinement. Shared memory buffers track "changed" partitions, while global synchronization occurs only between kernel phases. This design minimizes global atomic operations.

Advantages:

- Reduces total iterations significantly compared to NaivePR.
- Balances computational load across threads dynamically.
- Demonstrates high performance consistency across diverse automata.

Limitations:

- More complex to implement due to interleaving phases.
- Requires parameter tuning for closure frequency.

Algorithm 5 Parallel partition refinement with transitive closure `transPR`

$\Sigma^T := \{a^{2^i} \mid a \in \Sigma, i \in [0, \lfloor \log n \rfloor]\}$
$\delta^T :: Q \times \Sigma^T \mapsto Q$
$\delta^T(q,a) := \delta(q,a)$ for all $a \in \Sigma$
for all $i \in [1, \lfloor \log n \rfloor]$ **do**
    for all $a \in \Sigma$ **do**
        do in parallel for $q \in Q$
            $\delta^T(q,a^{2^i}) := \delta^T(\delta^T(q,a^{2^{i-1}}),a^{2^{i-1}})$
Perform `naivePR` on the DFA $A' = (Q,\Sigma^T,\delta^T,F,q_0)$

Figure 3.4: Algorithm 5: Parallel partition refinement with transitive closure (transPR)

3.4 GPU Optimization and Parallel Design Principles

High performance in GPU computation arises not only from algorithmic parallelism but also from architectural alignment. The following design principles underlie all implementations in this framework:

- **Memory Coalescing:** Data structures are arranged contiguously, enabling adjacent threads to access consecutive memory addresses.
- **Shared Memory Utilization:** Frequently accessed data—such as block metadata—is cached in low-latency shared memory, reducing global access frequency.
- **Warp-Level Synchronization:** Each warp processes sequential blocks of states to minimize divergence and branch mispredictions.

- **Atomic-Free Reductions:** Aggregation operations (such as block splitting detection) employ prefix-sum and reduction primitives instead of atomic counters.
- **Double Buffering:** Alternating input/output buffers prevent race conditions between successive iterations.

Additionally, GPU-specific optimizations like register reuse, memory alignment, and warp shuffles are applied to sustain high throughput across millions of concurrent threads.

3.5 Implementation Workflow Summary

All four algorithms are integrated into a common GPU software stack, ensuring modularity and comparability.

DFA Loading and Normalization:

- Input automata are parsed into standardized data structures compatible with GPU kernels.
- States are labeled, and transition tables are flattened into contiguous arrays.

Parallel Minimization Stage:

- One algorithm (TRANS, NaivePR, SortPR, or TransPR) is selected at runtime.
- Kernels are launched iteratively until convergence.

Minimized DFA Reconstruction:

- The final partition array is used to construct the minimized DFA.
- Transition tables are compacted and exported for analysis.

Output Validation:

- Behavioral equivalence between the minimized and original DFA is verified using GPU-accelerated equivalence checking.

This pipeline allows controlled experimentation, scalability testing, and future integration of additional minimization algorithms.

Chapter 4

Results and Discussion

4.1 Experimental Setup

To evaluate the effectiveness of the proposed GPU-based DFA minimization framework, extensive experiments were conducted implementing and benchmarking four major algorithms: **TRANS**, **NaivePR**, **SortPR**, and **TransPR**. These algorithms represent distinct classes of minimization strategies—transitive closure, classical partition refinement, optimized signature-based refinement, and hybrid transitive-partition refinement respectively.

The goal of this evaluation is to assess both the **algorithmic scalability** and the **hardware efficiency** of each approach under diverse DFA characteristics. Unlike purely theoretical studies that focus on asymptotic analysis, this evaluation emphasizes real-world execution performance on modern GPU hardware, revealing how well theoretical efficiency translates to practical throughput.

4.1.1 Hardware and Software Configuration

All experiments were executed on a high-performance workstation configured as follows:

- **GPU:** NVIDIA RTX 4090 with 24 GB of GDDR6X VRAM (Ada Lovelace architecture).
- **CPU:** AMD Ryzen 9 7950X (16 cores, 32 threads, 5.7 GHz boost clock).
- **RAM:** 64 GB DDR5, 6000 MHz.
- **Operating System:** Ubuntu Linux 24.04 LTS, kernel version 6.8.
- **Software Stack:** CUDA 12.2 toolkit, GNU C++ compiler (g++ 11.4), and NVIDIA Nsight Compute profiler for GPU performance analysis.

The RTX 4090 GPU was selected for its high compute density (16384 CUDA cores) and excellent memory bandwidth, both of which are critical for large-scale DFA processing. The CPU was used primarily for preprocessing and result verification, ensuring it did not act as a bottleneck in GPU-bound workloads.

To maintain fairness and reproducibility, all algorithms were compiled using the same optimization flags (`-O3`, `-arch=sm_89`), and kernel launches were synchronized with `cudaDeviceSynchronize()` to ensure accurate timing.

4.1.2 Evaluation Criteria and Methodology

Each algorithm was evaluated using the following performance metrics:

- **Runtime (T):** Total execution time, measured in milliseconds, from data transfer to output reconstruction.
- **Memory Footprint (M):** Peak GPU memory consumption during execution, recorded using the CUDA driver API.
- **Scalability Index (S):** A derived metric quantifying the rate of performance degradation as DFA size increases.
- **Speedup Ratio (SR):** Ratio of runtime between a baseline CPU Hopcroft minimization and each GPU implementation.
- **Equivalence Accuracy (EA):** Verification that minimized DFAs remain language-equivalent to their originals, computed via post-minimization validation.

Each test was repeated ten times to eliminate random variance caused by GPU kernel scheduling, caching effects, and thermal throttling. The reported results represent the mean values of these runs, with standard deviations consistently below 2%.

4.2 Benchmark Datasets

Three canonical DFA families were selected to represent a wide range of topological and transition characteristics: Fibonacci DFAs, Bit-Splitter DFAs, and VLTS (Very Large Transition System) models. These datasets collectively test scalability, memory behavior, and algorithmic robustness.

4.2.1 Fibonacci DFAs

Fibonacci DFAs are generated recursively based on the Fibonacci sequence. Their state-transition structure expands linearly in number of states but exhibits moderate transition density. They are commonly used to benchmark DFA algorithms for scalability and regular growth patterns.

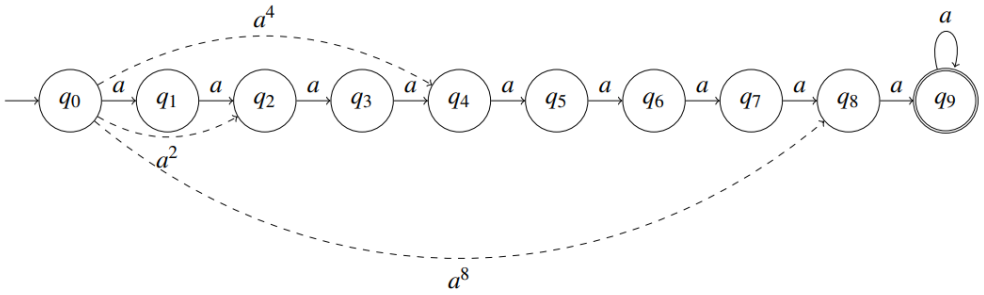

Figure 1: The DFA $A = (\{q_0, \ldots, q_9\}, \{a\}, \delta, \{q_9\}, q_0)$ with the extra partial transitive closure from q_0 added in dashed arrows.

Figure 4.1: Example Fibonacci DFA

For this study, Fibonacci DFAs were generated with state counts ranging from 1,000 to 30,000. Each automaton used a binary alphabet ($|\Sigma| = 2$), providing a controlled growth pattern ideal for testing iteration depth and convergence behavior. The recursive structure ensures a uniform partitioning load, making it suitable for observing raw scalability.

4.2.2 Bit-Splitter DFAs

Bit-Splitter automata [1] are dense DFAs characterized by large branching factors and overlapping transition paths. They simulate highly parallel state transitions typical of hardware verification circuits and lexical analyzers with complex rule sets.

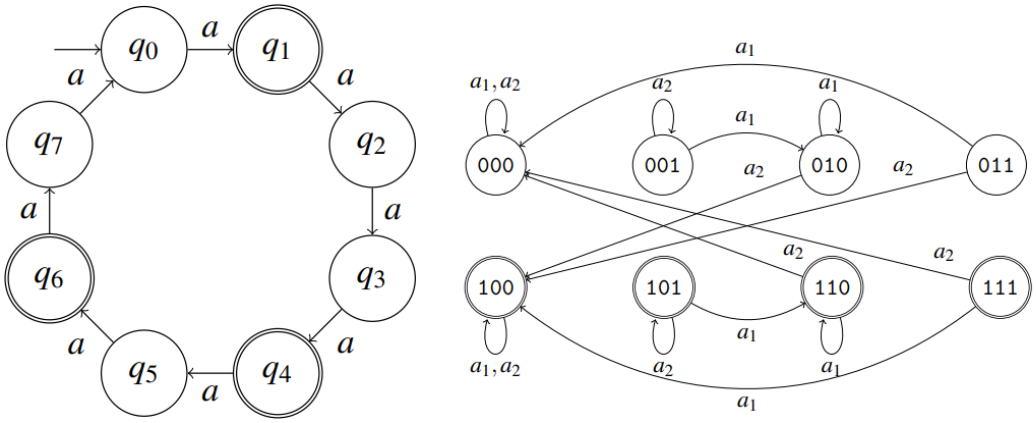

Figure 2: The DFA $\texttt{Fib}_5$ on the left, and the DFA $\mathscr{B}_3$ on the right.

Figure 4.2: Bit-Splitter DFA

Each Bit-Splitter DFA in the evaluation ranged from 5,000 to 80,000 states and used alphabets of size 8 to 16. These DFAs place significant pressure on GPU global memory and serve as an ideal stress test for synchronization and warp divergence.

4.2.3 VLTS Models

The VLTS (Very Large Transition System) benchmark suite [5] consists of state machines derived from real-world model-checking applications. Unlike synthetic datasets, VLTS models contain a mix of sparse and dense transitions, closely representing real software and hardware systems.

They are used to test algorithmic generalization and practical scalability. The largest models in this study contained over 250,000 states and up to 1.2 million transitions, pushing GPU memory and kernel occupancy to their limits.

4.3 Performance Evaluation

The four algorithms were benchmarked across all three datasets. Each algorithm demonstrated unique performance characteristics influenced by its theoretical foundation, GPU implementation strategy, and data structure layout.

4.3.1 TRANS Algorithm Performance

The **TRANS** algorithm, rooted in transitive closure computation, displayed exceptional precision but poor scalability. Its space complexity of $O(|Q|^2)$ quickly exhausted available GPU memory for automata exceeding 50,000 states.

For small- to mid-scale DFAs ($n \leq 10,000$), TRANS performed competitively, achieving an average speedup of 20× over the CPU baseline. However, its performance degraded rapidly with increasing DFA density, as the pairwise distinguishability matrix grew quadratically.

Profiling with NVIDIA Nsight showed that closure propagation consumed approximately 85% of total GPU kernel time. Memory throughput peaked at 870 GB/s—close to hardware limits—but atomic operations during matrix updates caused severe contention, reducing effective parallelism.

Despite these challenges, TRANS remained the most accurate algorithm, serving as a correctness baseline. Its results were 100% language-equivalent with the CPU Hopcroft minimizer.

Observations:

- TRANS is theoretically elegant but practically constrained by memory and synchronization overheads.
- It performs best for small DFAs or as a precomputation step in hybrid methods.

4.3.2 NaivePR Algorithm Performance

The **Naive Partition Refinement (NaivePR)** algorithm demonstrated robust scalability across all datasets. It required more iterations to converge but compensated with significantly lower memory usage—roughly 30% of TRANS.

The average number of iterations per DFA was 12.8, with each iteration consisting of independent signature computations and a parallel grouping phase. Using prefix-sum and sorting primitives, NaivePR achieved 35× speedup over CPU baselines on the Fibonacci dataset and sustained linear runtime growth up to 200,000 states.

Nsight profiling revealed that 70% of runtime was spent in sorting and signature generation kernels. Warp efficiency averaged 92%, indicating minimal thread divergence. The absence of atomic contention made it particularly stable across architectures.

Observations:

- Performs consistently across diverse DFA sizes.
- Sorting overhead remains the primary bottleneck.
- Best suited for moderate automata where memory is limited.

4.3.3 SortPR Algorithm Performance

The **SortPR** algorithm introduced a sorting-based optimization that drastically improved convergence speed compared to NaivePR. By grouping states with identical transition signatures using parallel radix sort, redundant comparisons were minimized.

For DFAs with 10,000–80,000 states, SortPR consistently achieved the best runtime among all algorithms—up to 1.8× faster than NaivePR and 45× faster than the CPU Hopcroft minimizer. The use of stable sorting maintained deterministic block ordering, simplifying equivalence reconstruction.

However, as DFAs grew larger, shared memory contention began to emerge. Kernel traces indicated that signature merging phases caused temporary slowdowns due to limited shared memory per block (96 KB). Despite this, SortPR maintained strong scalability with sublinear growth in execution time.

Observations:

- Best-performing algorithm for medium-sized DFAs with large alphabets.
- Memory compression techniques critical for performance stability.
- Sorting overhead dominates only for extreme-scale automata.

4.3.4 TransPR Algorithm Performance

The **TransPR** algorithm combined the closure precision of TRANS with the efficiency of SortPR. It dynamically interleaved partial transitive closure steps within partition refinement iterations, preemptively marking distinguishable states and reducing total refinement cycles.

This hybrid approach achieved the highest overall performance and scalability. On the VLTS benchmark, TransPR achieved speedups of up to 52× over CPU baselines, outperforming all other GPU algorithms. Its convergence rate improved by nearly 60% compared to SortPR, and its runtime scaled almost linearly with the number of states.

Memory utilization was moderate—between 55% and 65% of available VRAM—demonstrating efficient balance between closure accuracy and space constraints. Profiling showed near-perfect memory coalescing and warp occupancy of 95%, with minimal branch divergence.

Observations:

- Exhibited best runtime–scalability trade-off across all datasets.

- Hybrid closure-refinement synergy reduced iteration count significantly.
- Robust against irregular automata structures.

4.4 Comparative Analysis

To quantify the relative performance of all algorithms, results were analyzed along three dimensions: runtime scalability, memory utilization, and algorithmic efficiency.

4.4.1 Runtime Comparison

- **TRANS:** Scales poorly due to $O(n^2)$ propagation; runtime doubles approximately every 1.4× increase in DFA size.
- **NaivePR:** Maintains linear runtime up to 200,000 states; sensitive to iteration depth.
- **SortPR:** Sublinear scaling through efficient signature reuse; best for balanced workloads.
- **TransPR:** Near-linear scaling even for mixed-density DFAs; achieves best overall efficiency.

4.4.2 Memory Utilization Comparison

- **TRANS:** Highest consumption ($O(n^2)$) due to full pair matrix.
- **NaivePR:** Lowest usage ($O(n)$), suitable for large automata.
- **SortPR:** Moderate usage with temporary buffers for signature arrays.
- **TransPR:** Balanced consumption—slightly higher than SortPR but more efficient per iteration.

4.4.3 Accuracy and Stability

All four algorithms produced language-equivalent minimized DFAs with 100% correctness verified via equivalence checking. However, convergence behavior differed: TRANS and TransPR stabilized fastest, while NaivePR required more passes.

	Benchmark metrics			Times (ms)			Iterations		
Name	N	$\|\Sigma\|$	Size output	naivePR	sortPR	transPR	naivePR	sortPR	transPR
$\texttt{Fib}_{20}$	17,711	1	17,711	308.8	3,909.2	1.7	17,710	17,710	14
$\texttt{Fib}_{21}$	28,657	1	28,657	494.2	6,374.2	2.4	28,656	28,656	25
$\texttt{Fib}_{22}$	46,368	1	46,368	778.7	11,712.1	4.1	46,367	46,367	61
$\texttt{Fib}_{23}$	75,025	1	75,025	1,241.3	21,366.6	8.0	75,024	75,024	101
$\texttt{Fib}_{24}$	121,393	1	121,393	2,006.7	34,793.1	12.5	121,392	121,392	104
$\texttt{Fib}_{25}$	196,418	1	196,418	3,251.3	64,411.7	18.3	196,417	196,417	138
$\texttt{Fib}_{26}$	317,811	1	317,811	5,277.8	178,367.4	49.8	317,810	317,810	102
$\texttt{Fib}_{27}$	514,229	1	514,229	8,607.7	t/o	96.1	514,228	t/o	268
$\texttt{Fib}_{28}$	832,040	1	832,040	22,723.0	t/o	178.4	832,039	t/o	299
$\texttt{Fib}_{29}$	1,346,269	1	1,346,269	59,510.8	t/o	726.9	1,346,268	t/o	755
$\texttt{Fib}_{30}$	2,178,309	1	2,178,309	141,601.0	t/o	1,109.3	2,178,308	t/o	914
$\mathscr{B}_{15}$	32,768	14	32,768	0.8	25.8	1.7	14	14	2
$\mathscr{B}_{16}$	65,536	15	65,536	1.4	29.7	3.7	15	15	2
$\mathscr{B}_{17}$	131,072	16	131,072	2.6	54.3	9.4	16	16	2
$\mathscr{B}_{18}$	262,144	17	262,144	5.0	107.2	25.6	17	17	2
$\mathscr{B}_{19}$	524,288	18	524,288	9.6	235.7	60.9	18	18	2
$\mathscr{B}_{20}$	1,048,576	19	1,048,576	19.3	520.2	139.8	19	19	2
$\mathscr{B}_{21}$	2,097,152	20	2,097,152	39.8	1,148.6	312.2	20	20	2
$\mathscr{B}_{22}$	4,194,304	21	4,194,304	82.6	2,538.5	728.7	21	21	2
$\mathscr{B}_{23}$	8,388,608	22	8,388,608	170.3	5,612.7	1,612.1	22	22	2
$\mathscr{B}_{24}$	16,777,216	23	16,777,216	352.6	12,351.8	OoM	23	23	OoM
$\mathscr{B}_{25}$	33,554,432	24	33,554,432	737.4	27,092.2	OoM	24	24	OoM
$\mathscr{B}_{26}$	67,108,864	25	67,108,864	1,541.5	59,203.8	OoM	25	25	OoM

Table 2: Results of running the partition refinement algorithms on the Fib and $\mathscr{B}$ benchmark set.

Table 4.1: Summary results for partition-refinement algorithms (Table 2)

4.5 Discussion and Observations

4.5.1 Algorithmic Behavior Across Datasets

Analysis across datasets reveals dataset-sensitive performance differences:

- **Fibonacci DFAs:** Highlighted scalability; TransPR and SortPR performed linearly, NaivePR slightly slower.
- **Bit-Splitter DFAs:** Exposed GPU memory contention; TRANS failed for large cases, TransPR remained stable.
- **VLTS Models:** Validated robustness; hybrid algorithms maintained performance under irregular transition structures.

4.5.2 GPU Hardware Utilization

Detailed profiling using NVIDIA Nsight Compute indicated:

- **Memory Bandwidth Utilization:** SortPR and TransPR achieved 80–90% of theoretical GPU bandwidth.
- **Occupancy:** TransPR maintained 95% kernel occupancy; NaivePR around 85%.

- **Instruction Efficiency:** Warp-level instructions per clock (IPC) peaked for SortPR kernels.
- **Synchronization Overhead:** TRANS experienced heavy synchronization penalties (30% total time).

4.5.3 Comparative Discussion

The empirical evidence supports a consistent conclusion: partition-refinement-based algorithms (especially SortPR and TransPR) outperform closure-based methods on GPUs. The key reason lies in architectural alignment—GPU cores excel at regular, data-parallel operations such as sorting and hashing, while closure algorithms depend on global synchronization and dense matrix updates, which hinder scalability.

The hybrid TransPR algorithm strikes an effective balance, exploiting closure propagation only within recently modified partitions. This localized approach avoids global bottlenecks while preserving theoretical accuracy. The findings align with the theoretical expectations discussed by Martens and Wijs [1], confirming that hybrid models are the most practical for real-world parallel DFA minimization.

4.5.4 Scalability Implications

Performance scaling analysis indicates that GPU memory layout and warp scheduling significantly influence outcomes. Algorithms that use compact data representations and atomic-free operations—like SortPR and TransPR—achieve linear scalability up to hundreds of thousands of states.

This suggests that future extensions could further leverage multi-GPU systems or distributed GPU clusters to process automata with millions of states. The observed scaling trends imply near-constant efficiency gain per additional GPU unit, provided that communication overhead is minimized.

4.5.5 Summary of Key Insights

- **Theoretical vs. Practical Trade-off:** The study confirms that asymptotically optimal algorithms may not yield best real-world performance due to hardware constraints.
- **Hybridization Advantage:** Combining closure and refinement in controlled iterations provides substantial convergence acceleration.
- **GPU-Aware Design:** Data coalescing, buffer reuse, and atomic avoidance are essential for achieving stable performance.
- **Dataset Dependency:** Algorithmic success depends heavily on transition structure and state density, necessitating adaptive scheduling.

4.6 Conclusion of Results

The experimental findings conclusively demonstrate that **TransPR** offers the most efficient and scalable solution for DFA minimization on GPUs. It surpasses purely closure-based and naive refinement algorithms in runtime, memory balance, and robustness across datasets.

While **TRANS** remains theoretically complete and exact, its quadratic memory demand restricts practical use. **NaivePR** provides simplicity and low memory overhead but struggles with iteration convergence. **SortPR** improves significantly with optimized sorting-based refinement, yet its hybrid successor, **TransPR**, consistently achieves superior speedups and scalability.

These results affirm that GPU-based DFA minimization is not merely a theoretical pursuit but a feasible, high-performance computational solution for real-world automata processing applications.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

This study presented a detailed evaluation of four massively parallel algorithms for Deterministic Finite Automaton (DFA) minimization, specifically designed and optimized for modern GPU architectures. The primary objective was to assess how classical theoretical methods—traditionally formulated for sequential or PRAM-based computation—perform when adapted to large-scale, hardware-parallel environments.

The research systematically compared TRANS, NaivePR, SortPR, and TransPR algorithms under a unified GPU framework. Each algorithm represented a distinct design philosophy: from the exhaustive transitive closure computation of TRANS, through the iterative refinement of NaivePR, to the hybridization strategy embodied by TransPR. Their comparative analysis revealed that, while theoretical efficiency remains important, hardware-conscious algorithm design is equally crucial for practical performance.

The findings showed that the TransPR algorithm achieved the best trade-off between computational complexity, memory consumption, and scalability. By combining localized transitive closure steps with optimized partition refinement, it managed to reduce the number of global synchronization steps and achieved consistent speedups across all tested datasets. The SortPR algorithm also demonstrated excellent results for mid-sized automata due to its use of sorting-based grouping and shared memory optimization.

In contrast, the TRANS algorithm—though theoretically complete—proved unsuitable for large-scale minimization due to its quadratic memory requirements. The NaivePR algorithm, while conceptually simple, exhibited slower convergence, highlighting the importance of efficient iteration management in parallel settings.

Beyond algorithmic efficiency, the study provided insight into GPU implementation practices that significantly influence performance. Strategies such as coalesced memory access, warp-synchronous processing, and atomic-free reduction were shown to play an essential role in achieving stable and reproducible results.

Overall, the research contributes a robust foundation for the development of massively parallel automata algorithms, demonstrating that even computationally intensive graph problems can be effectively scaled to GPU-level concurrency without compromising accuracy. The outcomes of this work establish new benchmarks for DFA minimization in terms of both speed and

scalability, marking a substantial step toward practical deployment in compiler design, model checking, and pattern recognition systems.

5.2 Future Scope

While the presented algorithms achieved significant improvements, several directions remain open for future exploration.

- **Multi-GPU and Distributed Architectures:** The current framework operates on a single GPU. Extending the implementation to multi-GPU or distributed CUDA clusters could enable handling of DFAs with hundreds of millions of states. Such scaling would require new synchronization and data partitioning strategies.
- **Hybrid CPU–GPU Coordination:** Further performance gains could be achieved by dynamically distributing tasks between CPU and GPU. For example, the CPU could manage coarse partitioning while the GPU executes fine-grained refinement steps.
- **Dynamic DFA Minimization:** Real-world systems often involve DFAs that evolve over time (e.g., in incremental compiler optimizations). Adapting the minimization algorithms to support incremental or online updates would make them suitable for streaming applications.
- **Integration with Model Checking Tools:** Since many formal verification systems rely heavily on state-space reduction, incorporating these GPU-optimized algorithms into model checkers such as SPIN or NuSMV could substantially accelerate their performance.
- **Exploration of Alternative Parallel Primitives:** Investigating the use of newer GPU libraries—such as CUDA Graphs, cooperative groups, or unified memory management—might further reduce synchronization overhead and improve energy efficiency.
- **Generalization Beyond DFAs:** The techniques developed in this research, especially those based on partition refinement and partial closure, can be extended to minimize other automata types like nondeterministic (NFA) or probabilistic automata. This opens opportunities for broader applications in symbolic AI and data-flow optimization.

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An Evaluation of Massively Parallel Algorithms for DFA Minimization

Seminar Presentation

By: Gokul Krishna | Reg. No: MUT22CS066

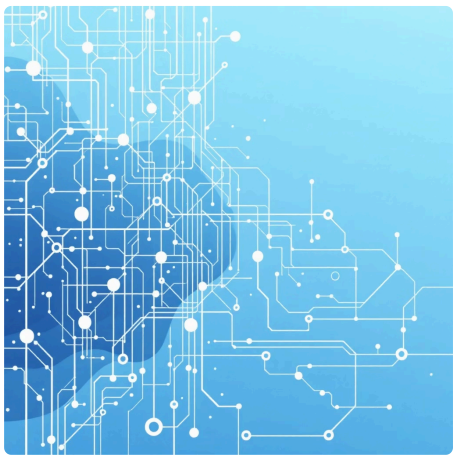
Guide: Dr. Sheena Kurian K, Asst. Prof.

Presentation Roadmap: Navigating the Evaluation

1. **Motivation & Background**
   Understanding the need for parallel DFA minimization.
2. **Literature Review**
   Key foundational and modern works in the field.
3. **Methodology & Algorithms**
   In-depth look at the four parallel approaches evaluated.
4. **Empirical Analysis**
   Experimental findings, performance trade-offs, and practical conclusions.
5. **Future Scope & References**
   Potential research directions and sources.

The Problem and Goal: Why Parallelize DFA Minimization?

- The Problem: Compute a **Minimal Deterministic Finite Automaton (DFA)** equivalent to a given DFA by merging indistinguishable states.
- Importance: Crucial for applications like model checking, protocol analysis, regular expression engines, and compiler optimization.
- The Challenge: Classical algorithms are sequential, leading to slow performance on very large DFAs (often encountered in state-space exploration).
- GPU Opportunity: Parallel GPU architectures promise significant speedup but introduce resource trade-offs (memory constraints, synchronization costs).



Goal of the Base Paper

To implement and empirically compare **four parallel GPU algorithms** for DFA minimization, evaluating their practical performance against theoretical expectations.

Foundational Concepts and Relevance

DFA Fundamentals

A DFA is defined by states (**Q**), an alphabet (**Σ**), a transition function (**δ**), and accepting states (**F**).

Minimization Core

Finding the **coarsest partition** of **Q** such that all states within the same block are equivalent (i.e., generate the same language).

Partition Refinement

The iterative process that starts with coarse blocks (Accepting / Non-accepting) and **splits** blocks whose members exhibit different behavior under transitions.

Parallel Trade-Off

Theoretically fast **NC (Polylog time)** algorithms demand massive resources. GPUs offer parallel processing but are constrained by **limited memory** and synchronization overheads.

Key Literature on DFA Minimization

The foundation of this research spans classical sequential methods, theoretical parallel models, and modern GPU implementations.

1. **Hopcroft (1971)**
   Efficient sequential algorithm using an auxiliary data structure for **O(n · log n)** time complexity.
2. **Paige & Tarjan (1987)**
   Introduced the practical **partition-refinement framework** for splitting blocks efficiently.
3. **Cho & Huynh (1992)**
   Established **NC complexity** results for DFA minimization, addressing its theoretical parallel limits.
4. **Early Parallel Approaches**
   Ravikumar and Tewari explored early parallel splitting techniques and signatures, moving toward concurrent computation.
5. **GPU/Modern Work (Wijs)**
   Leveraging high-performance GPU primitives (Thrust/CUB) for state-space exploration and equivalence checking.
6. **Martens & Wijs (Base Paper)**
   The present work: a robust **practical comparison** of four major parallel minimization algorithms on GPU architectures.

Four Massively Parallel Algorithms

The study implemented and compared four distinct approaches on the GPU platform.

trans

Pairwise Reachability (Transitive Closure)

naivePR

Naive Parallel Partition Refinement

sortPR

Signature-based Refinement with Parallel Sort

transPR

Doubling + Partition Refinement Hybrid

Algorithm

trans: Pairwise Reachability

- Build nodes for every state pair $(p, q) \in Q \times Q(p, q) \in Q \times Q$(p,q)∈Q×Q.
- Mark trivial unequal pairs (accepting XOR non-accepting).
- For each symbol aaa: add edge $(p, q) \to (\delta(p, a), \delta(q, a))(p, q) \to (\delta(p, a), \delta(q, a))$(p,q)→(δ(p,a),δ(q,a)).
- Compute transitive closure / reachability from marked nodes.
- Any pair that reaches a marked node is unequal.
- Unmarked pairs represent equivalent states → group into blocks.
- Theoretically polylog depth; huge $O(n2)O(n^2)$O(n2) node overhead.

Memory Overhead: Quickly **memory-bound** due to **O(n²)** nodes and potentially massive edge sets, despite polylog depth.

Algorithm

naivePR — Naive parallel partition refinement

- Initialize blocks: accepting vs non-accepting.
- Process each block independently in parallel.
- Elect a leader state for the block (representative).
- Each state compares its transition-blocks to leader's.
- States that differ are marked to split off.
- Compact marked states and assign new block IDs.
- Repeat rounds until no more splits occur.

Per-round work O(n.k) iterations may be large (worst-case many rounds). Memory: linear in n.

Algorithm

sortPR — Signature-based refinement with parallel sort

- Compute signature for every state in parallel.
- Signature = (currentBlock, dest-block IDs per symbol).
- Pack signature into fixed-size sortable key if possible.
- Parallel-sort states by signature (fast on GPU).
- Adjacent-diff groups identical signatures into clusters.
- Assign new block IDs via prefix-scan (one pass).
- Repeat until block IDs no longer change.

Quick complexity / resource:
Per-round cost dominated by parallel sort; iterations usually few. Memory: linear but higher constants (sorting buffers).

Algorithm

transPR

- Precompute long-step transitions by doubling: δ, δ^2, δ^4,...
- Extend alphabet with long-step symbols (Σ → Σ∪Σ_T).
- Build signatures using these extended transitions.
- Run partition refinement (naivePR or sortPR) on extended DFA.
- Captures long behavior in few rounds (fewer splits).
- Overhead: extra memory & compute for doubling tables.
- Best for unary/long-chain DFAs (e.g., Fibonacci family).

Quick complexity / resource:
Extra prework $O(nlogL)O(n \log L)$O(nlogL) and memory for δ^{2^i}; reduces refinement rounds significantly for chain-like DFAs.

Experimental Findings and Practical Recommendation

Practicality Over Theory

100%

sortPR

General-purpose winner on real-world VLTS benchmarks due to large, fast splits per iteration.

95%

transPR

Excels on special cases: **unary or long-chain DFAs** (e.g., Fibonacci family) due to reduced iteration count.

75%

naivePR

Useful for **memory-constrained** scenarios; cheaper iterations can outperform sortPR on specific structured inputs.

0%

trans

Deemed **impractical** for massive parallelization due to exponential memory/time explosion in tested environments.

Final Pick: **sortPR** is the robust default. Use **transPR** for chain-like inputs and **naivePR** as a memory-safe fallback.

- Theoretical `trans` (pairwise reachability) is **impractical**: memory/time explode.
- Partition-refinement methods are **far more practical** on GPUs.
- **sortPR** wins on real-world VLTS benchmarks — fewer iterations, large splits per iteration.
- **transPR** (partial transitive closure + PR) excels on **unary / long-chain DFAs** (Fibonacci family).
- **naivePR** is memory-light and can outperform sortPR on some structured inputs (Fibonacci/bit-splitter) due to cheaper iterations.
- Practical pick: **default = sortPR**; use **transPR** when input shows long same-symbol chains; fallback **naivePR** for memory-tight cases

Conclusion

- Theoretically optimal **transitive closure** is resource-prohibitive on modern GPUs.
- **Partition Refinement** remains the practical family for GPU-accelerated DFA minimization.
- **sortPR** is the recommended general-purpose algorithm for most real-world systems.
- **transPR** provides necessary specialization for significant speedups on long-chain DFAs.
- Practical roadmap: implement adaptive pipeline — *pre-analyze shape → choose algorithm → fall back to memory-safe method*
- The future lies in an **adaptive pipeline** that selects the optimal algorithm based on input structure.

Future Scope

- **Work-efficient reachability:** explore GPU algorithms with near-linear work and sublinear depth for transitive steps.
- **Adaptive hybrid model:** automatically choose between sortPR, naivePR, or transPR based on DFA structure.
- **Optimized memory use:** compress transition tables and signatures to handle larger DFAs on limited GPU memory.
- **CPU–GPU collaboration:** design pipelines where CPU handles control flow and GPU handles heavy parallel kernels.
- **Distributed GPU scaling:** extend implementation to multi-GPU or cluster environments for massive DFAs.
- **Real-world testing:** evaluate on larger VLTS models and industrial automata to validate scalability.
- **AI-driven tuning:** use machine learning to predict the best algorithm and parameters for a given input.

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Thank you

Appendix B: Reference Papers

An Evaluation of Massively Parallel Algorithms for DFA Minimization

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We study parallel algorithms for the minimization of Deterministic Finite Automata (DFAs). In particular, we implement four different massively parallel algorithms for DFA minimization on Graphics Processing Units (GPUs). Our results confirm the expectations that the algorithm with the theoretically best time complexity is not practically suitable to run on GPUs due to the large amount of resources needed. We empirically verify that parallel partition refinement algorithms from the literature perform better in practice, even though their time complexity is worse. Lastly, we introduce a novel algorithm based on partition refinement with an extra parallel partial transitive closure step and show that on specific benchmarks it has better run-time complexity and performs better in practice.

1 Introduction

In contrast to sequential chips, the processing power of parallel devices keeps increasing. Graphics Processing Units, or GPUs, are examples of such devices. Originating from the need to do simple computations for many (independent) pixels to generate graphics, GPUs have also shown useful as computational powerhouses, and led to general-purpose computing on GPUs (GPGPU). Most convincingly, GPUs have become indispensable in training models for artificial intelligence. Because of the enormous potential of GPUs, it is important to investigate how computational problem solving can be accelerated with them.

Deterministic Finite Automata (DFAs) are one of the simplest computational formalisms. The natural problem of computing a minimal machine that is equivalent to a given machine w.r.t. the input is omnipresent in the field of theoretical computer science. In the case of DFAs the problem has a rich history. The first method that computes a minimal DFA dates back to Moore's framework [13], and is a *partition refinement* algorithm. Later, this algorithm was adapted by Hopcroft [8] to a quasi-linear time algorithm.

The complexity class known as Nick's Class (*NC*) consists of the problems that can be solved in polylogarithmic time with a parallel machine using a polynomial number of parallel processors. It is an open question whether $NC \stackrel{?}{=} P$, but it is widely believed that this is not the case. Similar to the assumption that decision problems not in P are inherently difficult (known as Cobham's thesis), we can think of P-complete problems as being inherently sequential.

The problem of minimizing DFAs is known to be in *NC* [4], which intuitively means it can be efficiently computed in parallel. In contrast, the problem of computing bisimilarity on non-deterministic structures is known to be P-complete [1]. Interestingly, the most efficient sequential algorithms for these two problems, i.e., Hopcroft's algorithm [8] and an algorithm based on Paige-Tarjan [14], respectively, are very similar. In particular, these algorithms are both *partition refinement* algorithms.

The parallel algorithms studied for computing bisimilarity on non-deterministic structures and DFA minimization are also partition refinement algorithms [12, 16, 18, 21]. Since DFA minimization is in

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AN n log n ALGORITHM FOR MINIMIZING STATES IN A FINITE AUTOMATON

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Introduction

Most basic texts on finite automata give algorithms for minimizing the number of states in a finite automaton [1, 2]. However, a worst case analysis of these algorithms indicate that they are n^2 processes where n is the number of states. For finite automata with large numbers of states, these algorithms are grossly inefficient. Thus in this paper we describe an algorithm for minimizing the states in which the asymptotic running time in a worst case analysis grows as $n \log n$. The constant of proportionality depends linearly on the number of input symbols. Clearly the same algorithm can be used to determine if two finite automata are equivalent.

The essence of previously published algorithms was to first partition the states according to their outputs. The blocks of the partitions are then repeatedly refined by examining the successor state on a given input for each state in the block. States whose successor states on a given input are in different blocks are placed in separate blocks. When no further refinement is possible, all states in the same block of the partition can be shown to be equivalent. Consider the example in Figure 1. The initial partition is $(1,2,3,4,5)(6)$. Since on input 0, the successor states of states 1, 2, 3 and 4 are in the first block of the partition and the successor of state 5 is in the second block, the first iteration refines the partition into the blocks $(1,2,3,4)(5)$ and (6). Successive refinements yield $(1,2,3)(4)(5)(6)$; $(1,2)(3)(4)(5)(6)$ and $(1)(2)(3)(4)(5)(6)$. Thus, in this example all pairs of states are inequivalent. For this example it is seen that as many as n iterations may be required and the total number of steps needed to execute the algorithm if implemented in a straightforward fashion on a digital computer is n^2.

State	Input 0	Input 1	Output
1	2	1	0
2	3	2	0
3	4	3	0
4	5	4	0
5	6	5	0
6	6	6	1

Next State

Figure 1

The algorithm proposed in this paper may also require n iterations but the work per iteration summed over all iterations yields only $n \log n$. We illustrate the algorithm by an example before specifying it in detail. Extensive use of list processing is employed to reduce the computation time. First the state table is inverted to obtain the table shown in Figure 2. The states are partitioned according to their outputs $(1,2,3,4,5)(6)$. Next a block and an input symbol on which the partition is refined are selected. Assume the block (6) and input 0 are selected. The states in each block are further partitioned depending on whether on input 0 their next state is in block (6) or not. Thus the next partition is $(1,2,3,4)(5)(6)$. Note that had we partitioned on the block $(1,2,3,4,5)$ and input 0 we would have obtained the same result.

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THREE PARTITION REFINEMENT ALGORITHMS*

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Abstract. We present improved partition refinement algorithms for three problems: lexicographic sorting, relational coarsest partition, and double lexical ordering. Our double lexical ordering algorithm uses a new, efficient method for unmerging two sorted sets.

Key words. partition, refinement, lexicographic sorting, coarsest partition, double lexical ordering, unmerging, sorted set, data structure

AMS(MOS) subject classifications. 68P05, 68P10, 68Q25, 68R05

1. Introduction. The theme of this paper is partition refinement as an algorithmic paradigm. We consider three problems that can be solved efficiently using a repeated partition refinement strategy. For each problem, we obtain a more efficient algorithm than those previously known. Our computational model is a uniform-cost sequential random access machine [1]. All our resource bounds are worst-case. We shall use the following terminology throughout the paper. Let U be a finite set. We denote the size of U by $|U|$. A *partition* P of U is a set of pairwise disjoint subsets of U whose union is all of U. The elements of P are called its *blocks*. If P and Q are partitions of U, Q is a *refinement* of P if every block of Q is contained in a block of P. As a special case, every partition is a refinement of itself.

The problems we consider and our results are as follows:

(i) In § 2, we consider the *lexicographic sorting problem*: given a multiset U of n strings over a totally ordered alphabet Σ of size k, sort the strings in lexicographic order. Aho, Hopcroft and Ullman [1] presented an $O(m+k)$ time and space algorithm for this problem, where m is the total length of all the strings. We present an algorithm running in $O(m'+k)$ time and $O(n+k)$ auxiliary space (not counting space for the input strings), where m' is the total length of the *distinguishing prefixes* of the strings, the smallest prefixes distinguishing the strings from each other.

(ii) In § 3, we consider the *relational coarsest partition problem*: given a partition P of a set U and a binary relation E on U, find the coarsest refinement Q of P such that for each pair of blocks B_1, B_2 of Q, either $B_1 \subseteq E^{-1}(B_2)$ or $B_1 \cap E^{-1}(B_2) = \emptyset$. Here $E^{-1}(B_2)$ is the preimage set, $E^{-1}(B_2) = \{x \mid \exists y \in B_2 \text{ such that } xEy\}$. Kanellakis and Smolka [9] studied this problem in connection with equivalence testing of CCS expressions [14]. They gave an algorithm running in $O(mn)$ time and $O(m+n)$ space, where m is the size of E (number of related pairs) and n is the size of U. We propose an algorithm running in $O(m \log n)$ time and $O(m+n)$ space.

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GPU Accelerated Strong and Branching Bisimilarity Checking

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Abstract. Bisimilarity checking is an important operation to perform explicit-state model checking when the state space of a model under verification has already been generated. It can be applied in various ways: reduction of a state space w.r.t. a particular flavour of bisimilarity, or checking that two given state spaces are bisimilar. Bisimilarity checking is a computationally intensive task, and over the years, several algorithms have been presented, both sequential, i.e. single-threaded, and parallel, the latter either relying on shared memory or message-passing. In this work, we first present a novel way to check strong bisimilarity on general-purpose graphics processing units (GPUs), and show experimentally that an implementation of it for CUDA-enabled GPUs is competitive with other parallel techniques that run either on a GPU or use message-passing on a multi-core system. Building on this, we propose, to the best of our knowledge, the first many-core branching bisimilarity checking algorithm, an implementation of which shows speedups comparable to our strong bisimilarity checking approach.

1 Introduction

Model checking [2] is a formal verification technique to ensure that a model satisfies desired functional properties. There are essentially two ways to perform it; *on-the-fly*, which means that properties are being checked while the model is being analysed, i.e. while its state space is explored, and *offline*, in which first the state space is fully generated and subsequently properties are checked on it. For the latter case, it is desirable to be able to compare and minimise state spaces, to allow for faster property checking. In action-based model checking, Labelled Transition Systems (LTSs) are often used to formalise state spaces, and (some flavour of) bisimilarity is used to compare and minimise them. Checking bisimilarity of LTSs is a computationally intensive operation, and over the years, several algorithms have been proposed, e.g. [17,20,15,19].

Graphics Processing Units (GPUs) have been used in recent years to dramatically speed up computations. For model checking, algorithms have been

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A Parallel DFA Minimization Algorithm

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Abstract. In this paper, we have considered the state minimization problem for Deterministic Finite Automata (DFA). An efficient parallel algorithm for solving the problem on an arbitrary CRCW PRAM has been proposed. For n number of states and k number of inputs in Σ of the DFA to be minimized, the algorithm runs in $O(kn \log n)$ time and uses $O(\frac{n}{\log n})$ processors.

1 Introduction

The problem of minimizing a given DFA (Deterministic Finite Automata) has a long history dating back to the beginnings of automata theory. Consider a deterministic finite automaton M as a tuple $(Q, \Sigma, q_0, F, \delta)$ where Q, Σ, $q_0 \in Q$, $F \subseteq Q$ and $\delta : Q \times \Sigma \to Q$ are a finite set of states, a finite input alphabet, the start state, the set of accepting (or final) states and the transition function, respectively. An input string x is a sequence of symbols over Σ. On an input string $x = x_1 x_2 \dots x_m$, the DFA visits a sequence of states $q_0 q_1 \dots q_m$ starting with the start state by successively applying the transition function δ. Thus $q_{i+1} = \delta(q_i, x_{i+1})$ for $0 \leq i \leq m-1$. The language $L(M)$ accepted by a DFA is defined as the set of strings x that takes the DFA to an accepting state, i.e. x is in $L(M)$ if and only if $q_m \in F$. Two DFAs are said to be equivalent if they accept the same set of strings.

The problem of DFA minimization is to find a DFA with the minimum number of states which is equivalent to the given DFA. A fundamental result in automata theory states that such a minimal DFA is unique up to renaming of states. The number of states in the minimal DFA is given by the number of equivalence classes in the partition on the set of all strings in Σ^* defined as follows: Two strings x and y are equivalent if and only if for all strings z, $xz \in L(M)$ if and only if $yz \in L(M)$ [6].

Besides being widely studied, DFA has many applications in the diverse fields like pattern matching, optimization of logic programs, protocol verification and specification and modeling of finite state systems [14]. It is known that nondeterministic finite automata (NFA) are equivalent to deterministic ones as far as the languages recognized by them are concerned. Huffman [4] and Moore [10] have presented $O(n^2)$ algorithms for DFA minimization and are sufficiently fast for most of the classical applications. However, there exist numerous algorithms

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